Zach J. Patterson

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PROFESSIONAL EXPERIENCE

Case Western Reserve University — **Assistant Professor**
Mechanical and Aerospace Engineering, — 2025 - Present
PI, Cybernetics and Physical Intelligence Lab — Cleveland, OH

Massachusetts Institute of Technology — **Postdoctoral Associate**
CSAIL — 2022 - 2024

TU Delft — **Visiting Researcher**
Dept. of Cognitive Robotics — July - Aug 2023

EDUCATION

PhD Mechanical Engineering — **Carnegie Mellon University**
Dissertation title: *Soft Robots for Simulator Validation, Control, and Comparative Biomechanics* — July 11, 2022
Dissertation Committee: *Carmel Majidi (Chair), Henry Astley, Sarah Bergbreiter, Robert Wood*

B.S. Mechanical Engineering — **University of Pittsburgh**
December 2017

JOURNAL PUBLICATIONS

For the most up to date list, see Google Scholar (Link).

9. *Zach J. Patterson, Henry C. Astley, Carmel Majidi, "Soft robotic brittle star shows the influence of mass distribution on underwater walking." Bioinspiration & Biomimetics, March 2025. doi:10.1088/1748-3190/adbecb*
8. *Akua K. Dickson, Juan C. P. Garcia, Meredith L. Anderson, Ran Jing, Sarah Alizadeh-Shabdiz, Audrey X. Wang, Charles DeLorey, Zach J. Patterson, Andrew P. Sabelhaus, "Safe Autonomous Environmental Contact for Soft Robots Using Control Barrier Functions." IEEE Robotics and Automation Letters, 2025. doi:10.1109/lra.2025.3609669*
7. *Andrew P. Sabelhaus, Zach J. Patterson, Anthony T. Wertz, Carmel Majidi, "Safe Supervisory Control of Soft Robot Actuators." Soft Robotics, August 2024. doi:10.1089/soro.2022.0131*
6. *Zach J. Patterson, Dinesh K. Patel, Sarah Bergbreiter, Lining Yao, Carmel Majidi, "A Method for 3D Printing and Rapid Prototyping of Fieldable Untethered Soft Robots." Soft Robotics, April 2023. doi:10.1089/soro.2022.0003*
5. *Richard Desatnik, Zach J. Patterson, Przemysław Gorzelak, Samuel Zamora, Philip LeDuc, Carmel Majidi, "Soft robotics informs how an early echinoderm moved." Proceedings of the National Academy of Sciences, November 2023. doi:10.1073/pnas.2306580120*
4. *Xiaonan Huang, Zach J. Patterson, Andrew P. Sabelhaus, Weicheng Huang, Kiyn Chin, Zhijian Ren, Mohammad K. Jawed, Carmel Majidi, "Design and Closed�Loop Motion Planning of an Untethered Swimming Soft Robot Using 2D Discrete Elastic Rods Simulations." Advanced Intelligent Systems, September 2022. doi:10.1002/aisy.202200163*
3. *Zach J. Patterson, Andrew P. Sabelhaus, Carmel Majidi, "Robust Control of a Multi-Axis Shape Memory Alloy-Driven Soft Manipulator." IEEE Robotics and Automation Letters, April 2022. doi:10.1109/lra.2022.3143256*
2. *Richard Dauksher, Zachary Patterson, Carmel Majidi, "Characterization and Analysis of a Flexural Shape Memory Alloy Actuator." Actuators, August 2021. doi:10.3390/act10080202*
1. *Xiaonan Huang, Michael Ford, Zach J. Patterson, Masoud Zarepoor, Chengfeng Pan, Carmel Majidi, "Shape memory materials for electrically-powered soft machines." Journal of Materials Chemistry B, 2020. doi:10.1039/d0tb00392a*

CONFERENCE PUBLICATIONS

6. *Makram Chahine, T. K. Rusch, Zach J. Patterson, Daniela Rus, "Improving Efficiency of Sampling-based Motion Planning via Message-Passing Monte Carlo." CORL, September 2025*
5. *Zach J. Patterson, Emily R. Sologuren, Daniela Rus, "Design of Trimmed Helicoid Soft-Rigid Hybrid Robots." 2025 IEEE 8th International Conference on Soft Robotics (RoboSoft), April 2025. doi:10.1109/robosoft63089.2025.11020854*
4. *Zach J. Patterson, Cosimo D. Santina, Daniela Rus, "Modeling and Control of Intrinsically Elasticity Coupled Soft-Rigid Robots." 2024 IEEE International Conference on Robotics and Automation (ICRA), May 2024. doi:10.1109/icra57147.2024.10610229*
3. *Zach J. Patterson, Wei Xiao, Emily Sologuren, Daniela Rus, "Safe Control for Soft-Rigid Robots with Self-Contact Using Control Barrier Functions." 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft), April 2024. doi:10.1109/robosoft60065.2024.10522000*
2. *James M. Bern, Zach J. Patterson, Leonardo Z. Yañez, Kristoff K. Misquitta, Daniela Rus, "A Fabrication and Simulation Recipe for Untethering Soft-Rigid Robots with Cable-Driven Stiffness Modulation." 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), October 2023. doi:10.1109/iros55552.2023.10341630*
1. *Xiaonan Huang, Weicheng Huang, Zachary Patterson, Zhijian Ren, M. K. Jawed, Carmel Majidi, "Numerical Simulation of an Untethered Omni-Directional Star-Shaped Swimming Robot." 2021 IEEE International Conference on Robotics and Automation (ICRA), May 2021. doi:10.1109/icra48506.2021.9561301*
0. *Zach J. Patterson, Andrew P. Sabelhaus, Keene Chin, Tess Hellebrekers, Carmel Majidi, "An Untethered Brittle Star-Inspired Soft Robot for Closed-Loop Underwater Locomotion." 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), October 2020. doi:10.1109/iros45743.2020.9341008*

UNDER REVIEW + PRE-PRINTS

3. *Maximilian Stölzle, T. K. Rusch, Zach J. Patterson, Rodrigo Pérez-Dattari, Francesco Stella, Josie Hughes, Cosimo Della Santina, Daniela Rus, "Learning to Move in Rhythm: Task-Conditioned Motion Policies with Orbital Stability Guarantees." arXiv preprint arXiv:ARXIV.2507.10602, 2025. doi:10.48550/ARXIV.2507.10602*
2. *Zach J. Patterson, Emily Sologuren, Cosimo Della Santina, Daniela Rus, "Design and Control of Modular Soft-Rigid Hybrid Manipulators with Self-Contact." arXiv preprint arXiv:ARXIV.2408.09275, 2024. doi:10.48550/ARXIV.2408.09275*

IN-PREPARATION

TEXTBOOK CHAPTERS

1. *W Huang, Z.J. Patterson, C. Majidi, M.K. Jawed, "Modeling Soft Swimming Robots using Discrete Elastic Rod Method." Bioinspired Sensing, Actuation, and Control in Underwater Soft Robotic Systems, Nov. 2020.*

INVITED TALKS + PRESENTATIONS

13. *Chair, Robot Locomotion Session, RoboSoft 2025.*
12. *Design and Control of Soft-Rigid Hybrid Robots, Harvard Microrobotics Lab Seminar, Harvard, July 2024.*
11. *Blending Soft and Rigid Structures for Physical Intelligence in Robotics, Mechanical Engineering Departmental Seminar, Case Western, April 2024.*
10. *Blending Soft and Rigid Structures for Physical Intelligence in Robotics, Mechanical Engineering Departmental Seminar, Delaware, April 2024.*
9. *Blending Soft and Rigid Structures for Physical Intelligence in Robotics, Mechanical Engineering Departmental Seminar, UIUC, March 2024.*
8. *Blending Soft and Rigid Structures for Physical Intelligence in Robotics, Mechanical Engineering Seminar, University of Southern California, Feb. 2024.*
7. *Blending Soft and Rigid Structures for Physical Intelligence in Robotics, Mechanical Engineering Departmental Seminar, University of Tennessee, Feb. 2024.*
6. *Physical Intelligence and Control for Soft Robotics, Soft Robotics Course, University of Michigan, Feb. 2024.*

5. *Towards Interaction Safe Soft-Rigid Hybrid Robots Action Lab Seminar, Northeastern University, Dec. 2023.*
4. *Understanding Embodied Intelligence with Comparative Biomechanics and Control Theory Mechanical Engineering Department Seminar, UMass Amherst, Sep. 2023.*
3. *"Understanding, Modeling, and Controlling Embodied Intelligence." EPFL Create Lab, EPFL, July 2023.*
2. *"Soft Robot Simulation and Control." Soft Robots: Mechanics, Design and Modeling Course, Carnegie Mellon, March 2022.*
1. *"Biomimetic Approaches to Robot Design ." Conversations on Bioinspired Materials, UC Berkeley, Feb. 2021.*

REVIEWER FOR JOURNALS AND CONFERENCES

- *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*
- *IEEE International Conference on Robotics and Automation (ICRA)*
- *IEEE Transactions on Robotics (T-RO)*
- *IEEE Transactions on Automation Science and Engineering (T-ASE)*
- *IEEE Robotics and Automation Letters (RAL)*
- *IEEE International Conference on Soft Robotics (RoboSoft)*
- *Nature Scientific Reports*
- *Soft Robotics*

TEACHING

- INSTRUCTOR *Case Western Reserve University*

  Fall 2025 | EMAE 285, Mechanical Engineering Measurements
  - Undergraduate level course on applied probability and statistics, measurements, data analysis, and machine learning.
- INSTRUCTOR *Case Western Reserve University*

  Spring 2025 | EMAE 485, Nonlinear Dynamics and Control
  - Graduate level course in control theory blending the classical and the modern with a focus on robotics. See the course website for more details: `https://nonlinear-control-emae-485.github.io/`.
  - 4.71/5.0 avg evaluation
- GRADUATE STUDENT INSTRUCTOR (GSI). *Carnegie Mellon University*

  Fall 2019 | ME 24351, Dynamics
  Fall 2021 | ME 24352, Dynamic Systems and Controls

MEDIA

- WALL STREET JOURNAL - *'Forget Humanoids. At MIT, Worms and Turtles Are Inspiring a New Generation of Robots.' - 2025*
- BBC - *'How soon will robots be able to behave like humans?' - 2024*
- MASHABLE - *'MIT is developing a soft robot that takes its inspiration from sea turtles' - 2023*
- SCIENTIFIC AMERICAN - *'Robotics 'Revives' a Long-Extinct Starfish Ancestor' - 2023*
- HACKADAY - *'Underwater Crawling Soft Robot Stays in Shape' - 2020*
- SYFY WIRE - *'Meet PATRICK, Carnegie Mellon's New Spongebob-Inspired Robotic Starfish' - 2020*
- TECHXPLORE - *'This is PATRICK: Meet the brittle star-inspired robot that can crawl underwater' - 2020*